

Session 5

Recall = The poles of $G(s)$ are the ^{roots/}zeros of $1 + G(s)$

stability $\rightarrow$ If $1 + G(s)$ has a zero in the RHP $G(s)$ goes to infinity.
If not $\checkmark$.

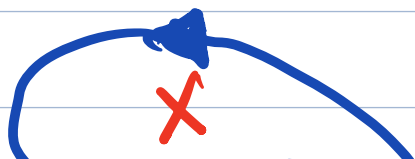
How to determine this??

We use Nyquist plot to analyze stability!

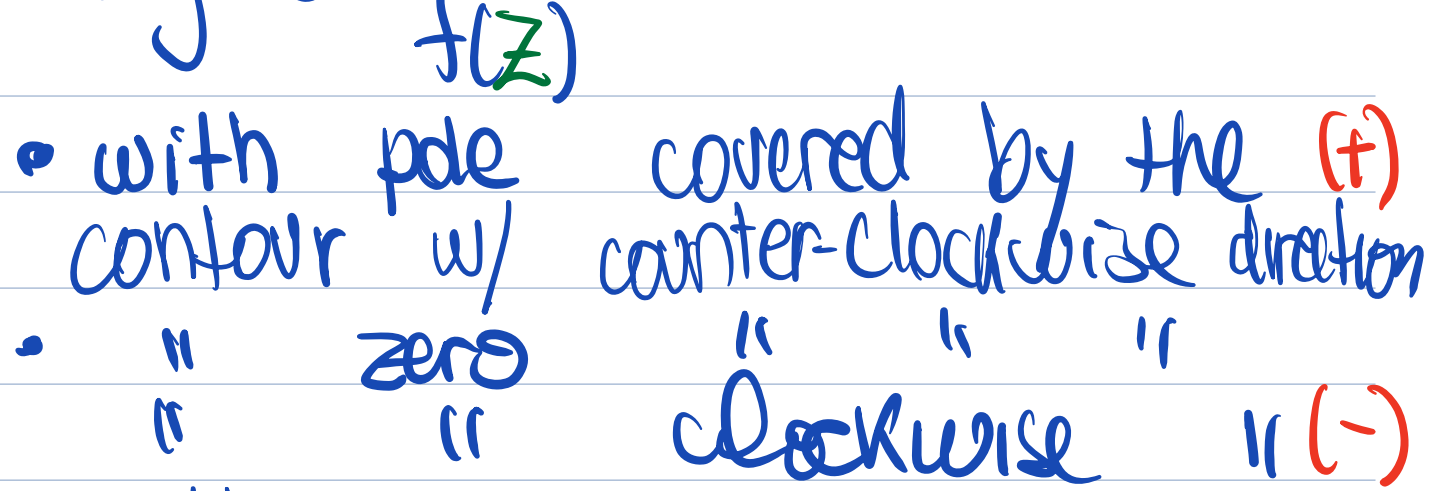
But first we need to know:

① Cauchy's argument principle

contour
integral $\oint$



$$\frac{1}{2\pi i} \oint \frac{f'(z)}{f(z)} dz = \#Z - \#P$$



The diagram illustrates the relationship between the s-plane and the Nyquist plot. On the left, the s-plane is shown with a horizontal real axis and a vertical imaginary axis. A red circle labeled 'o' is in the upper half-plane. Two green 'x' marks are on the negative real axis. An orange curve, labeled 'Nyquist contour' and $A(s)$, starts at the origin, goes into the right half-plane, and returns. An arrow points from this contour to the right, where a Nyquist plot is shown in the complex plane. The Nyquist plot is an orange curve that encircles the origin twice, with the label 'Nyquist plot' and 'Im' above it.

0

mapping
from s to w
through the trans.
for function

Argument Principle

of
r

A contour map a complex function
will encircle the origin $Z-P$ times

Don't write this...

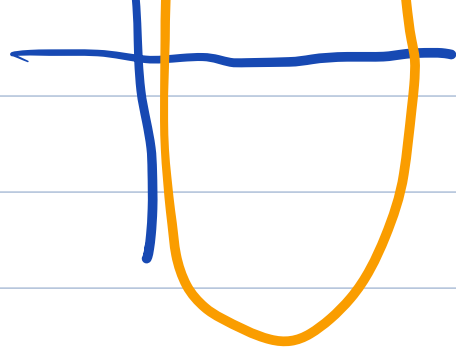
$\therefore$ You can tell the relative difference between the
poles and # zeros inside of a contour
by how many time the plot circles
the origin and in which direction

s -plane

w -plane

Ex

What
can you tell



$\Rightarrow$

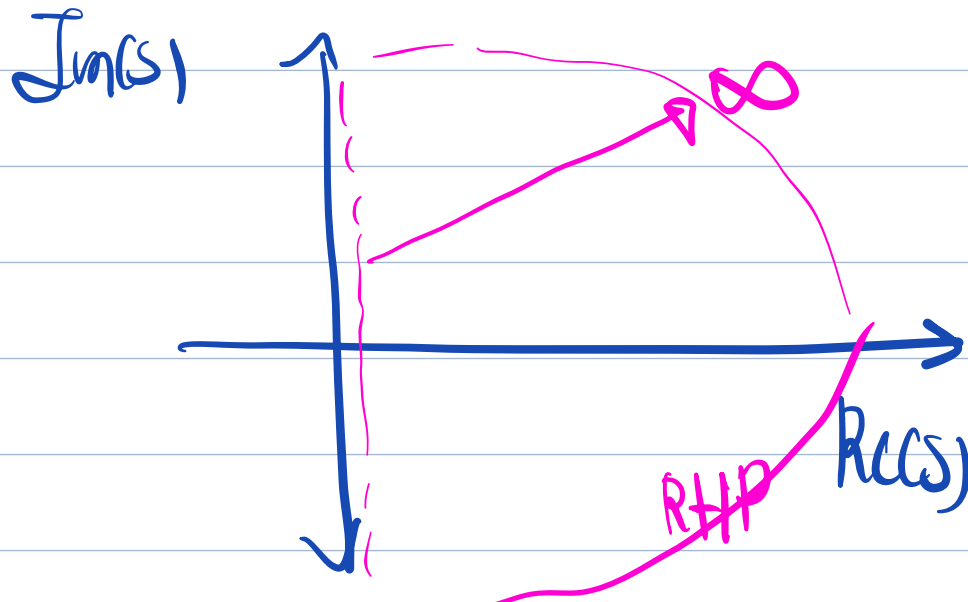


about this?

= It circle the origin
once in CW direction. Answer: One more zero than pole!
2 more poles than zero

Now, coming back to Nyquist...

Q: Does $1 + G(s)$ have zeros in the RHP?



- 1) + we draw a contour in the whole RHP
- 2) Draw the Nyquist plot
- 3) The number of times that the plot of $1 + G(s)$ encircles

clear the origin corresponds to $z-p$:

- From (3)
- 4) $\Rightarrow$ THE NUMBER OF TIMES THAT THE plot of G_{OL} encircles -1 corresponds to its $z-p$.

We shift the origin to -1 and we $\rightarrow$ shift -1 (From origin to -1)
Nyquist criterion $\rightarrow$ get $N = Z - P_{OL}$
 $N + P_{OL} = P_{CL}$

Poles of the G_{CL} in RHP = P of the G_{OL} in RHP
+ # of encirclements of G_{OL} around -1

number of poles in the close loop in RHP $\rightarrow$

$$P_{CL} = P_{OL} + N$$

$\rightarrow$ no of encirclements of -1
 $\rightarrow$ number of open loop poles

For the system to be stable:

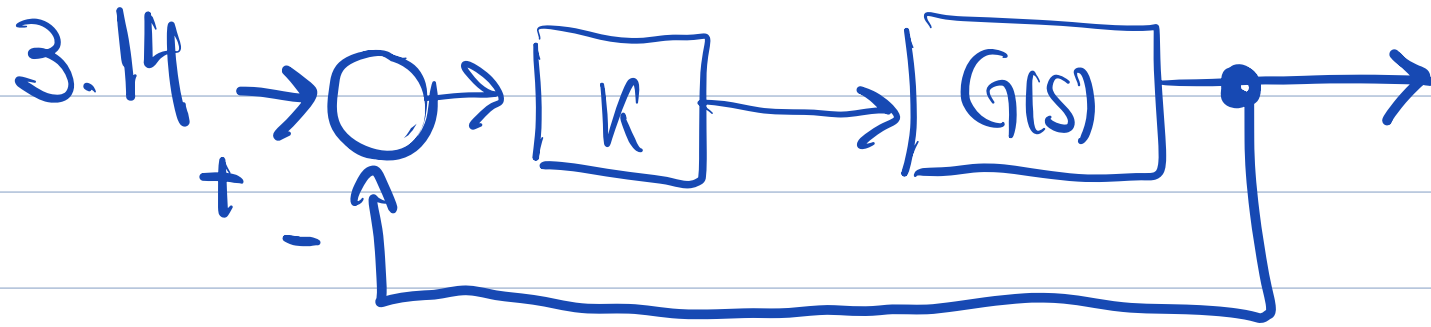
$$P_{cl} = 0 = P_{ol} + N \Rightarrow P_{ol} = -N$$

How to draw a Nyquist plot? (we plot $G_{ol}(s)$)

- 1) $s = j\omega$ ($G_{ol}(s)$)
 - 2) draw from $\omega = 0$ to $\omega = \infty$
 - 3) Symmetry against the Real axis. (c.c.w +, c.w -)
- REMEMBER THE DIRECTION !!**

Role of K :

K just scales the diagrams!!! (in Root Locus it was not like this, REMEMBER)



For $K = 1$

$$\frac{Y(s)}{R(s)} = \frac{G(s) \cdot K}{1 + G(s) \cdot K}$$

(a)

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

$$P_{CL} = P_{OL} + N$$

For the system to be stable $P_{OL} + N = 0$

$$N = -P_{OL}$$

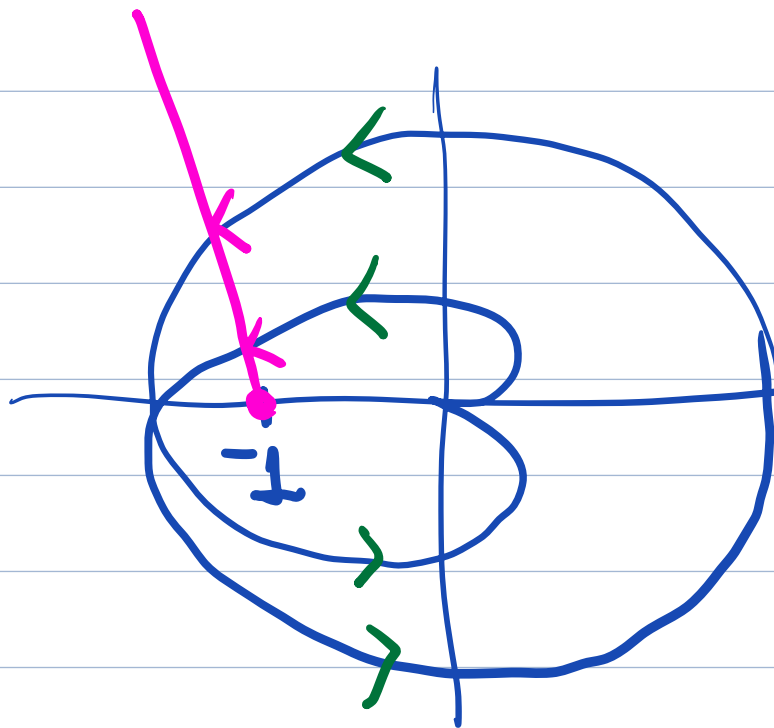
$$\text{Open loop} = K G(s) = G(s)$$

+ They say $P_{OL} = 0$ so the system to be stable $N = 0$ (no encirclements around -1)

i) -2 not encircle $\Rightarrow N = 0$ STABLE ✓

ii) $\Rightarrow N = 0$ STABLE ✓

iii)



The encirclements are passing through the lmp in the same direction

$$N = \pm 2$$

REMEMBER

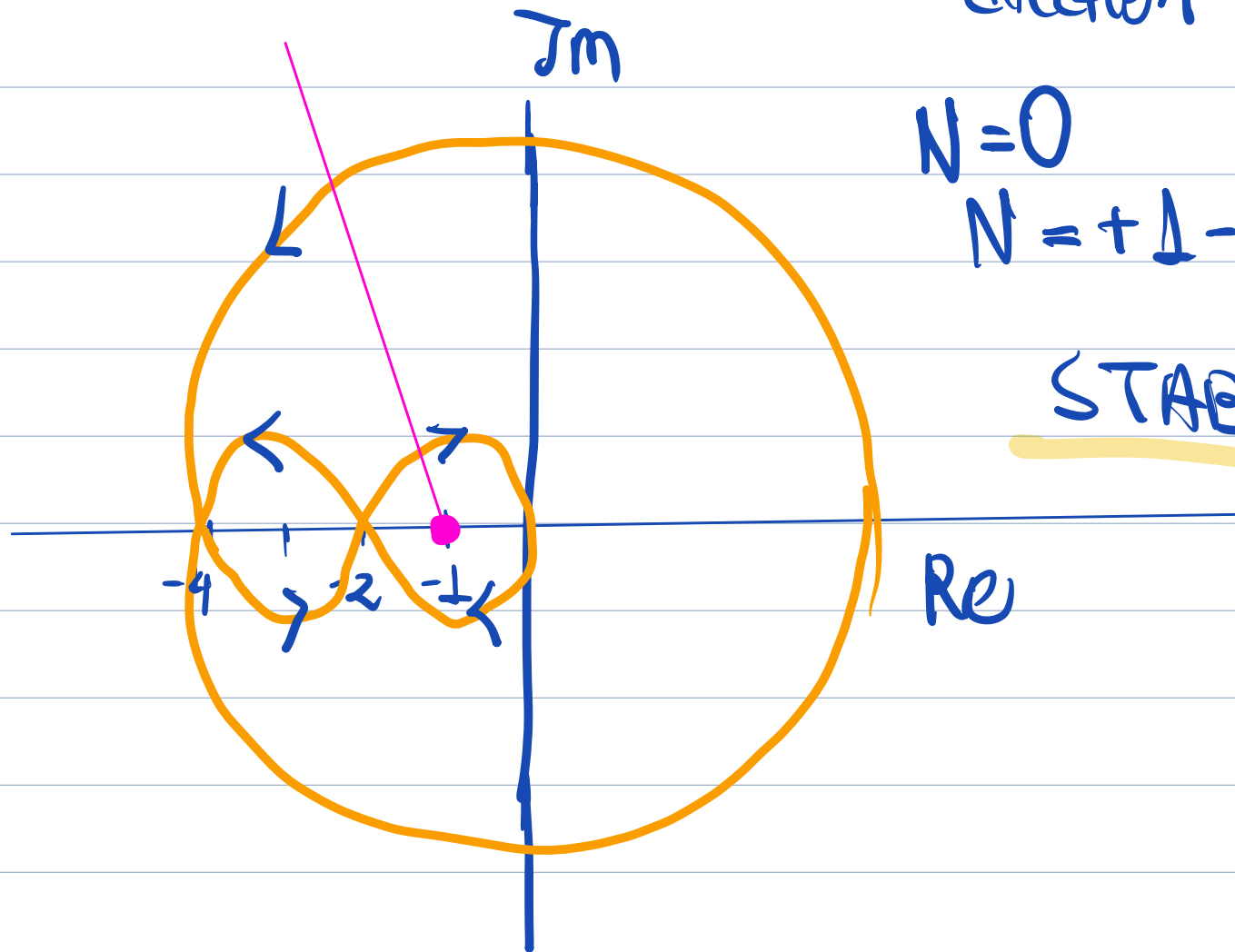
+2 C.C.W

-2 C.W

UNSTABLE

→ we don't know this direction

(iv)



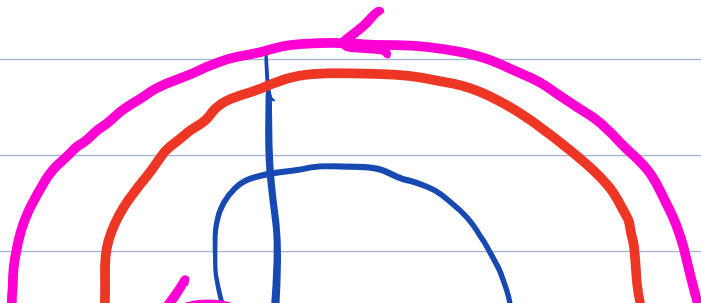
$$N=0$$

$$N = +1 - 1 = 0 //$$

STABLE

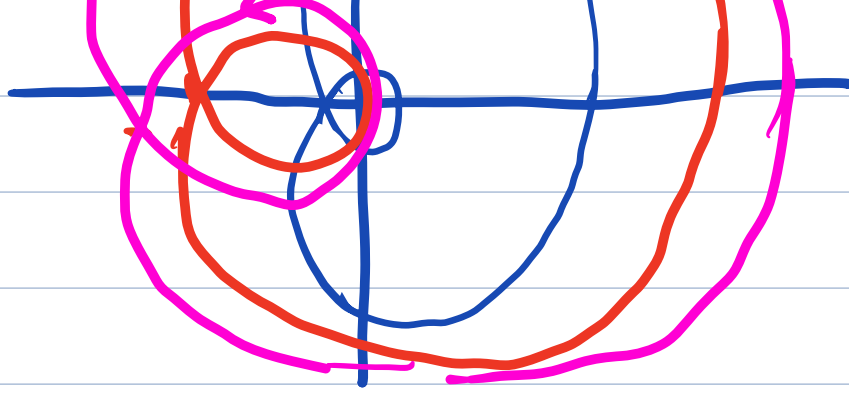
(b) If $K > 0$ (Before $K=1$)

i)



is stable until $K=?$

point of interest



$K=1 \rightarrow -0.4$
 Use Rule of 3 (rule of 3)

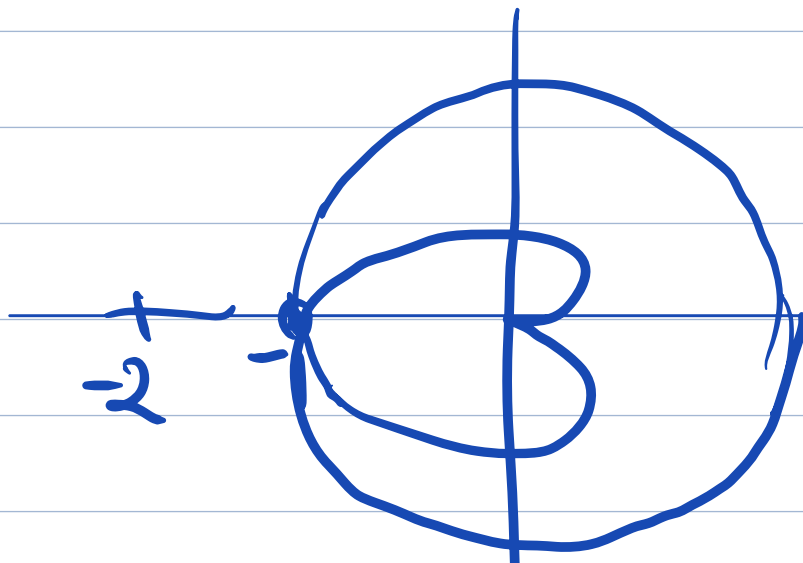
$$K=1 \rightarrow -0.4$$

$$K = \frac{-1}{-0.4} = 2.5$$

For $K > 2.5$ $N = \pm 2 \Rightarrow$ unstable

(i)
(ii)

Always stable $\forall K$



point interval

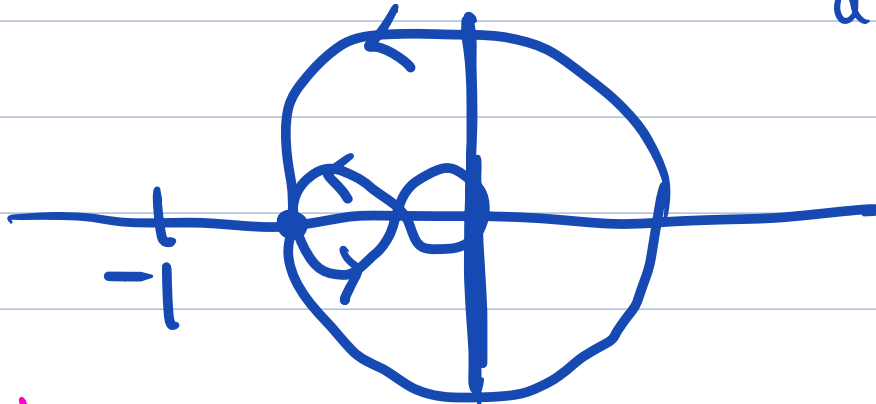
$$K=1 \rightarrow -2$$

$$K=? \rightarrow -1$$

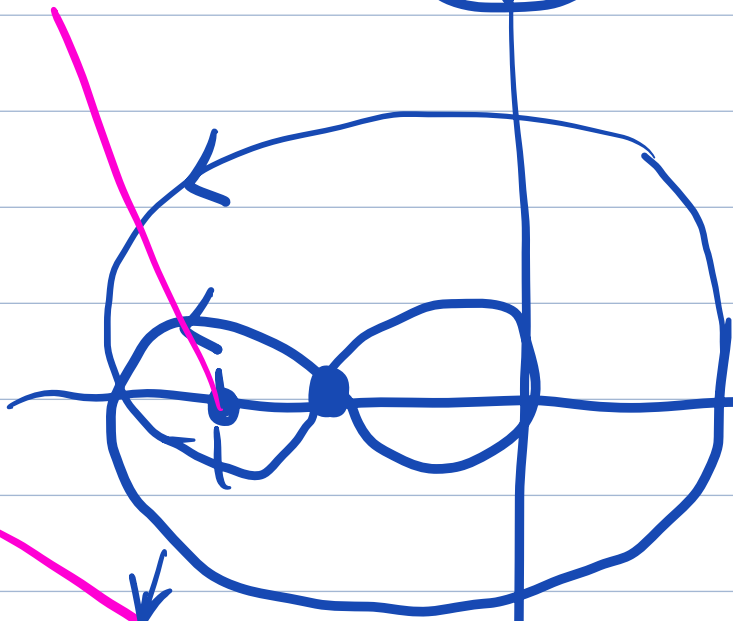
$$K = 0.5$$

So for $k < 0.5$ it's stable

iv)



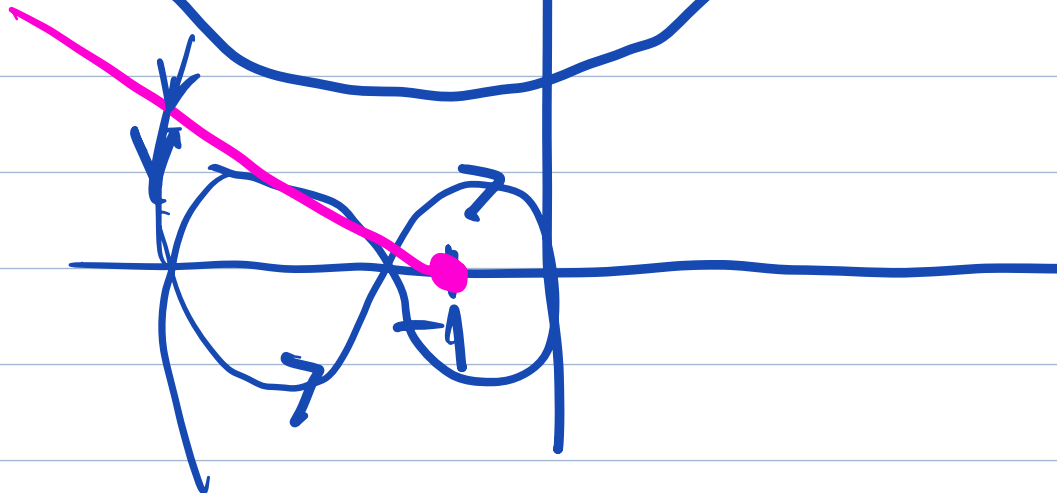
a) STABLE if
 $K=1$ Inter -4
 $K=$ -1
 $K < 0.25$ ✓



b) UNSTABLE if

$K=1$ -2
 $K=$ -1

$K = 0.5$



Stable $K < 0.25$
 $K > 0.5$

3.15

a) Draw $G(s) = \frac{1}{s}$

$$s = j\omega$$

Magnitude

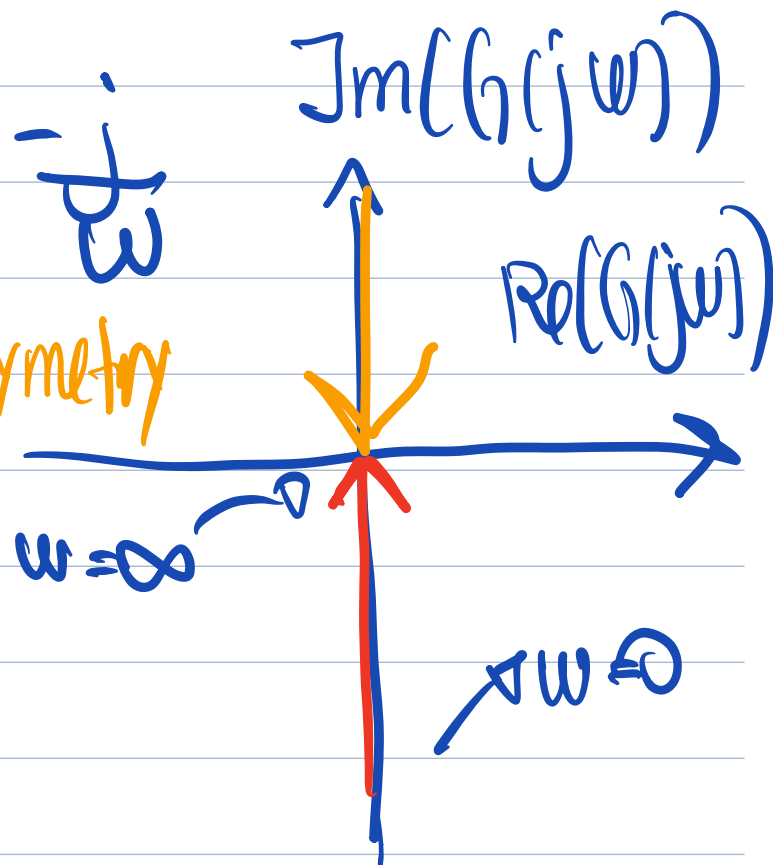
$$\left| \frac{1}{\omega} \right|$$

Angle

$$= -\frac{\pi}{2}$$

$$G(j\omega) = \frac{1}{j\omega} = -\frac{j}{\omega}$$

symmetry

b) Draw $G(s) = \frac{1}{s^2}$

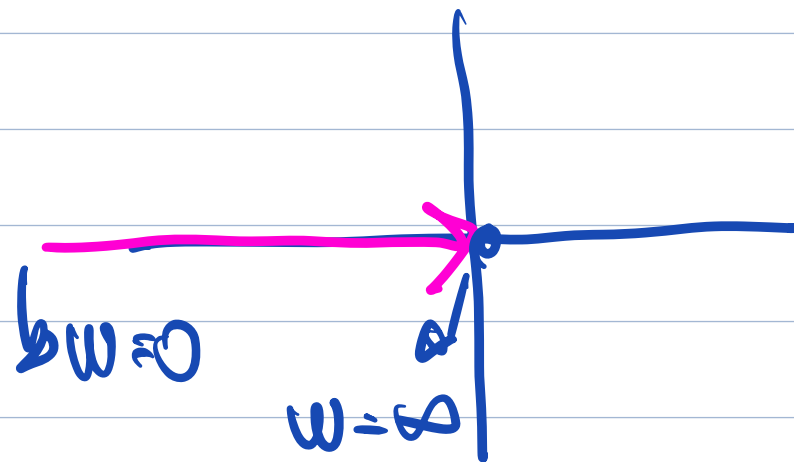
$$s = j\omega$$

$$G(j\omega) = -\frac{1}{\omega^2}$$

Magnitude

$$\left| \frac{1}{\omega^2} \right|$$

$$\text{angle} = \pi$$



3.16

a) $G(s) = P$
 $K=1$

$$G_d = K \cdot G(s)$$

$$G_{cl} = \frac{K G(s)}{1 + K G(s)}$$

We know $G(s) \rightarrow$ no poles RHP

$$P_{OL} = 0$$

From $P_{cl} = N + P_{OL}$

$$\omega=0 \quad N=0$$

condition for stability.

intercept

a) $K=1$ -1.5
 $K = -1$

$$K = \frac{1}{1.5} = \frac{2}{3}$$

$$N = -2$$

-1.5

-4

$\omega=10$

$\omega=\infty$

Unstable for
 $K=1$ with
 $N=-2$

$\omega=2$

-3

$K < \frac{2}{3}$ to be stable //

b) steady state error, as a function of K



$$\frac{E(s)}{R(s)} = \frac{1}{1+FG}$$
$$E(s) = \frac{R(s)}{1+FG}$$

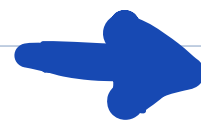
$$\lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \frac{1}{1+KG(s)}$$

$$R(s) - E(s) \cdot FG = E(s) \Rightarrow \lim_{s \rightarrow 0} \frac{1}{1+KG(s)} = \lim_{\omega \rightarrow 0} \frac{1}{1+KG(j\omega)}$$

$$R(s) = (1+FG)E(s)$$

$$\frac{E(s)}{R(s)} = \frac{1}{1+FG} = \frac{1}{1+KG(0)} = \frac{1}{1+2K} // \text{ for } K < \frac{2}{3}$$

$$c) \textcircled{1} G_{cl} = \frac{\frac{K}{s} \cdot G(s)}{1 + \frac{K}{s} G(s)}$$



$$1 + \underbrace{\frac{K}{s} G(s)}$$

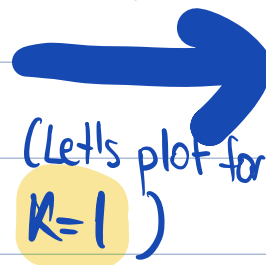
New $G_{ol}(s)!!$

$\textcircled{2}$

For Nyquist

$$G_{ol}(s) = \frac{K \cdot G(s)}{s}$$

$$s = j\omega$$



$$\frac{G(s)}{s} \Big|_{s=j\omega} = \frac{G(j\omega)}{j\omega}$$

$$\text{To plot} = \frac{-G(j\omega)}{\omega} j$$

What are the changes on $G(s)$??

Magnitude:

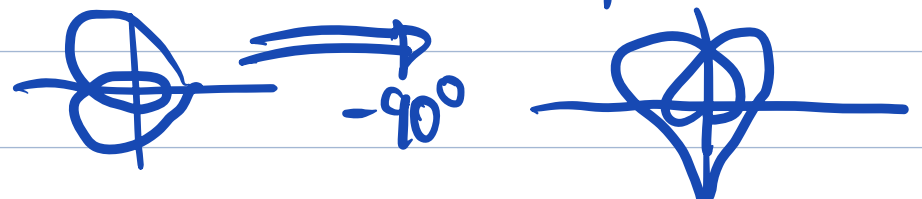
$$\left| \frac{G(j\omega)}{\omega} \right|$$

Angle:

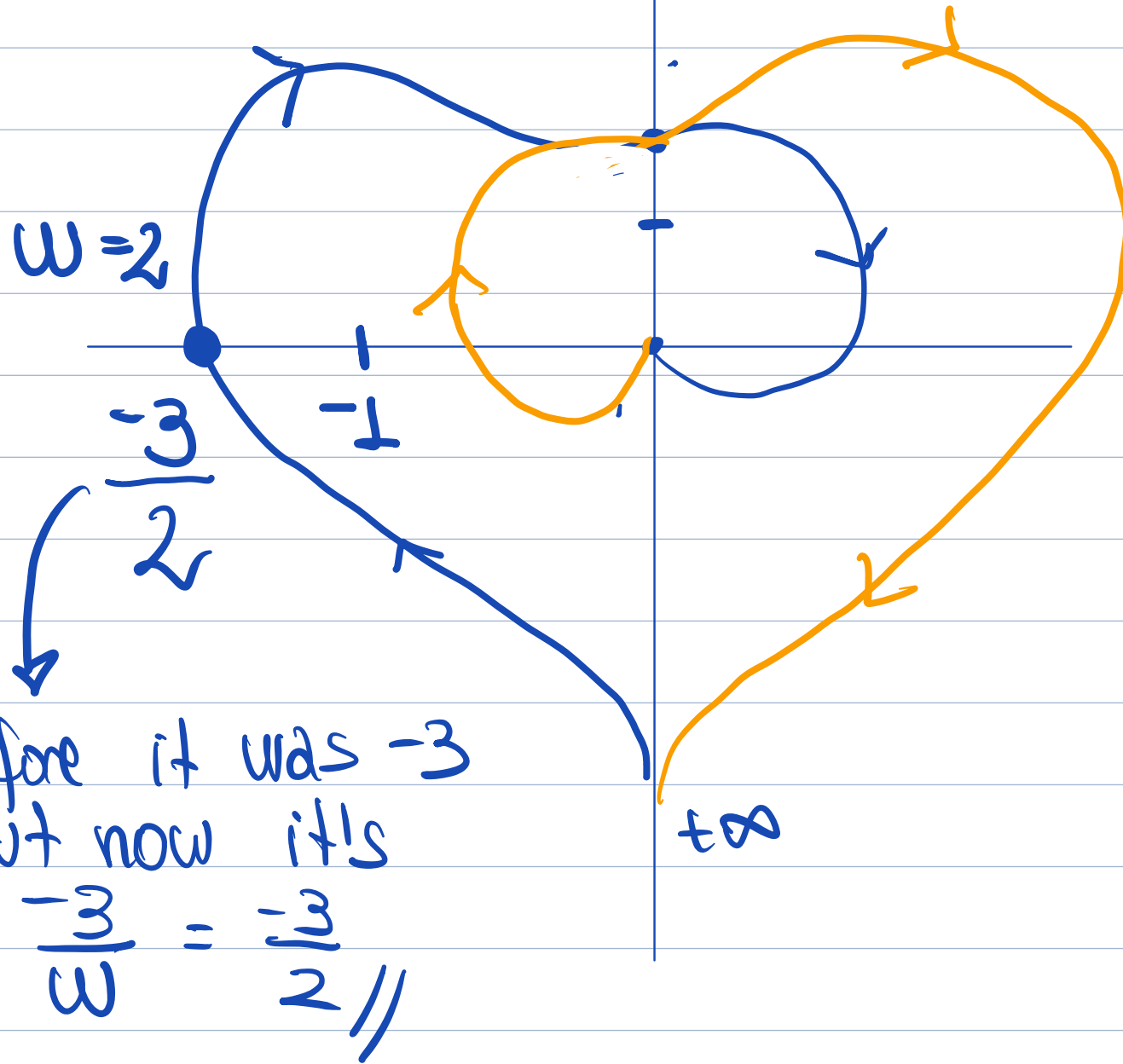
$$\angle G(j\omega) - 90$$

↳ we have to divide everything by $|\omega|!!$

↳ We rotate the plot in $-\frac{\pi}{2}!$



Now we plot for
 $K=2$ with the new
 Go!!



Rule of 3: new part
 of interest
 $K=1$ $-\frac{3}{2}$
 $K=$ -1

$$K < \frac{2}{3}$$